

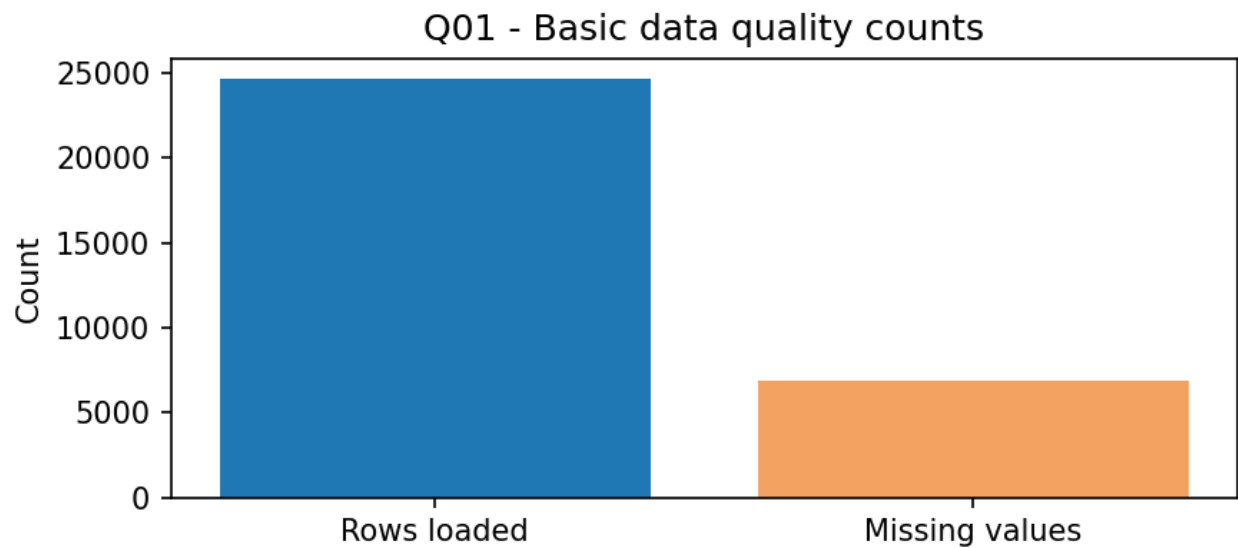
Supply Chain Review

Data quality review

Q01 Are the source tables complete and internally consistent?

At a glance

24,617 total rows and 6,927 missing values.



What the chart shows

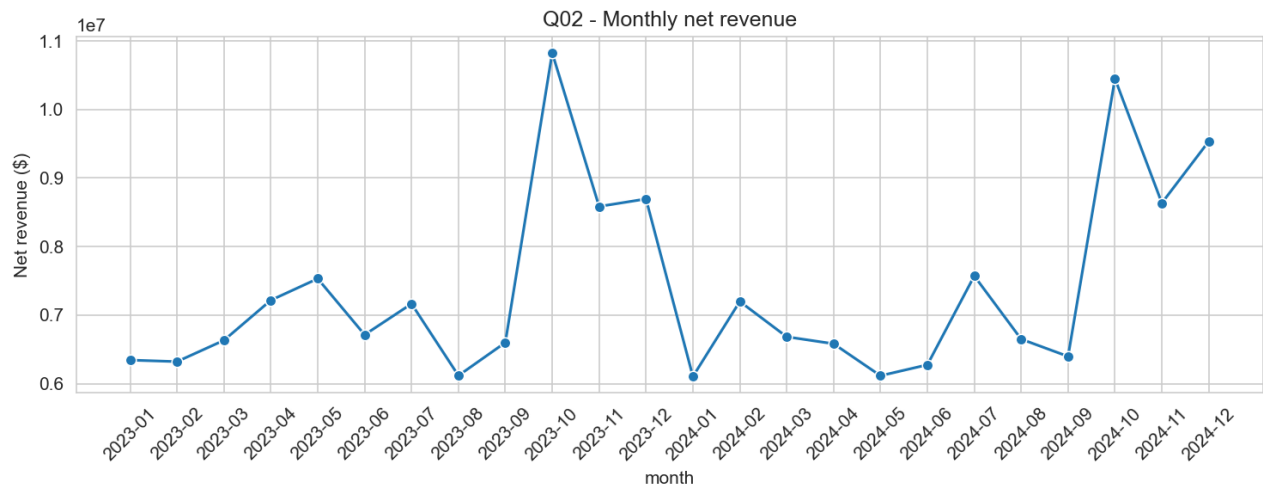
The project contains ten related CSV tables covering customers, dates, facilities, products, suppliers, inventory, procurement, production, sales, and shipments. The missing-value count is a starting point for review rather than proof that the data is unusable.

What this means: Before publishing new KPIs, document which missing values are expected, especially blank shipment delay reasons for shipments that were not delayed.

Monthly sales trend

Q02 How does revenue change by month?

At a glance
October 2023 was the highest month at \$10,825,244.



What the chart shows

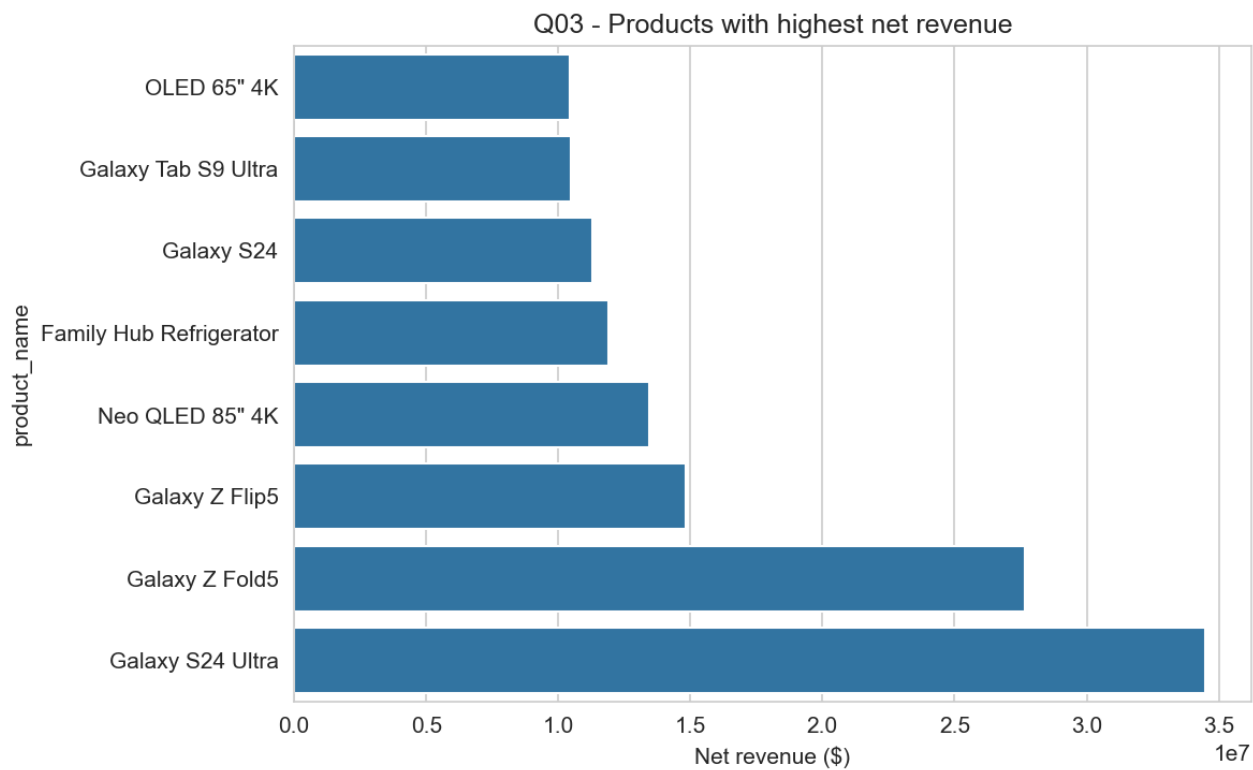
The line chart shows how net revenue moved across the 2023–2024 period. The peak identifies when demand or sales activity was strongest, while lower points can be reviewed against campaigns, product launches, or supply availability.

What this means: Use the monthly pattern to support inventory and production planning, but investigate the business reason for the peak before using it as a forecast.

Product revenue contribution

Q03 Which products generate the most net revenue?

At a glance
Galaxy S24 Ultra led with \$34,489,109 in net revenue.



What the chart shows

The bar chart compares the highest-revenue products. Revenue leadership shows where sales volume and price are contributing most to the business, but it should be read alongside margin and inventory exposure.

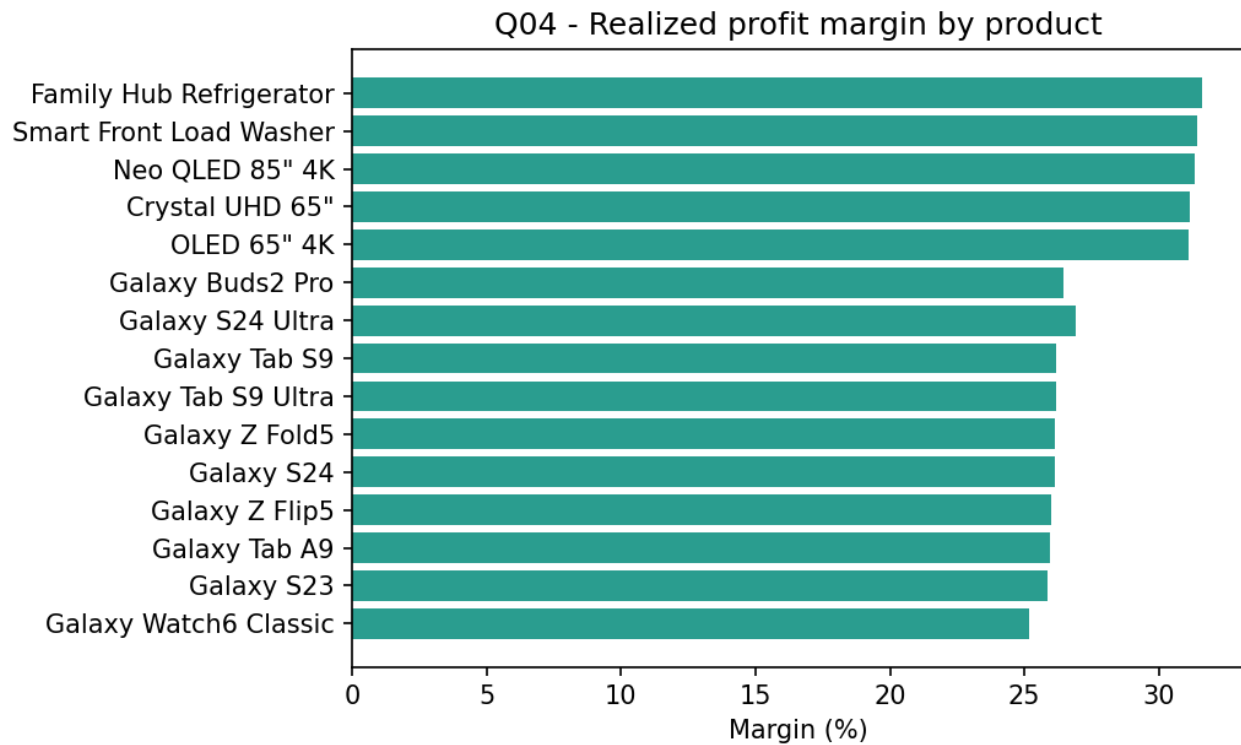
What this means: Protect availability for high-revenue products while checking that their margin and stock position justify priority allocation.

Product profitability

Q04 Which products have the strongest realized profit margins?

At a glance

Family Hub Refrigerator led with a 31.61% realized margin.



What the chart shows

The chart compares product-level profit economics after the recorded sales costs and discounts. A product with a smaller revenue total can still be attractive if each sale produces a stronger margin.

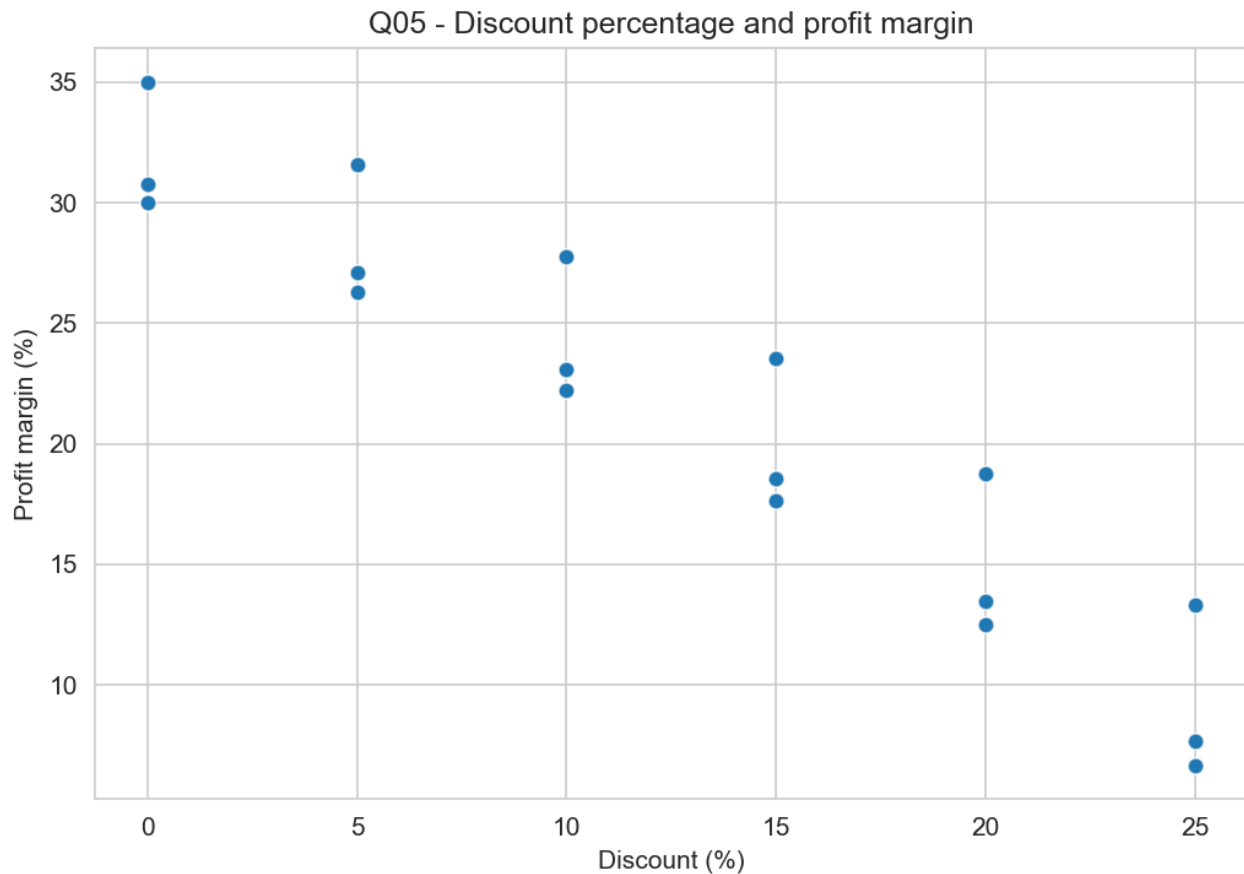
What this means: Use margin and revenue together when setting product priorities; do not allocate inventory based on revenue alone.

Discount and margin relationship

Q05 Is heavier discounting associated with lower profit margin?

At a glance

The correlation was **-0.91**; higher discounts were linked to lower margins.



What the chart shows

The scatter plot shows a strong downward relationship between discount percentage and profit margin in the sales records. This is an association in the data, not proof that every discount causes the same margin change.

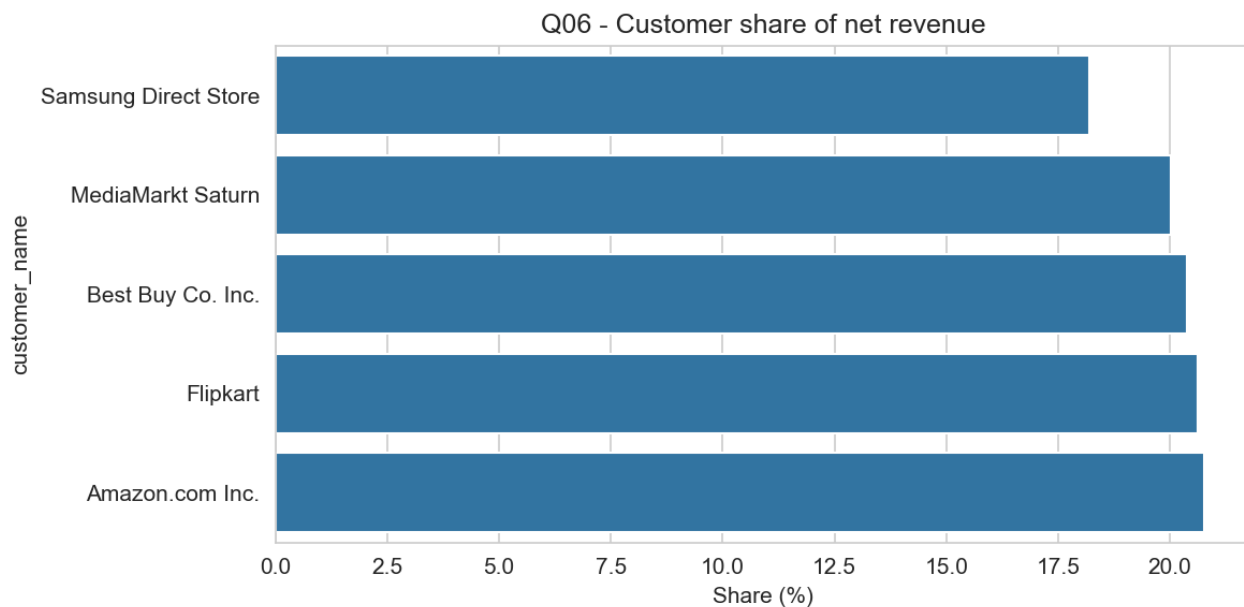
What this means: Review the 6–10% and 11%+ discount bands and require a clear commercial reason when discounting reduces profit economics.

Customer concentration

Q06 Is revenue concentrated among a small number of customers?

At a glance

The top two customers generated 41.39% of net revenue.



What the chart shows

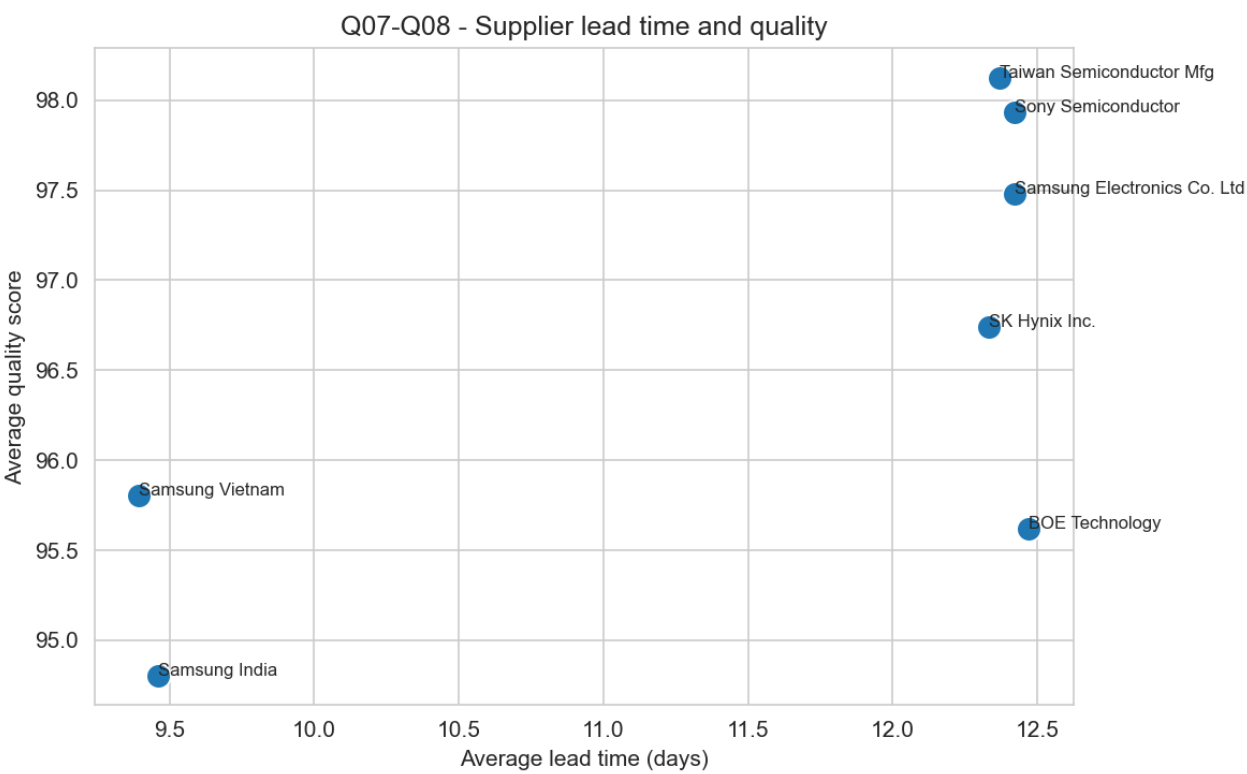
The customer-share chart shows how much of total net revenue comes from each account. A concentrated customer base can support large volumes, but it also creates exposure if one account changes its buying pattern.

What this means: Track the top accounts closely and continue developing smaller channels to reduce dependency risk.

Supplier lead time

Q07 Does procurement lead time vary by supplier?

At a glance
Samsung Vietnam was fastest at 9.39 average days.



What the chart shows

The supplier chart compares average procurement lead times. Differences matter because longer replenishment cycles can require earlier ordering or more safety stock.

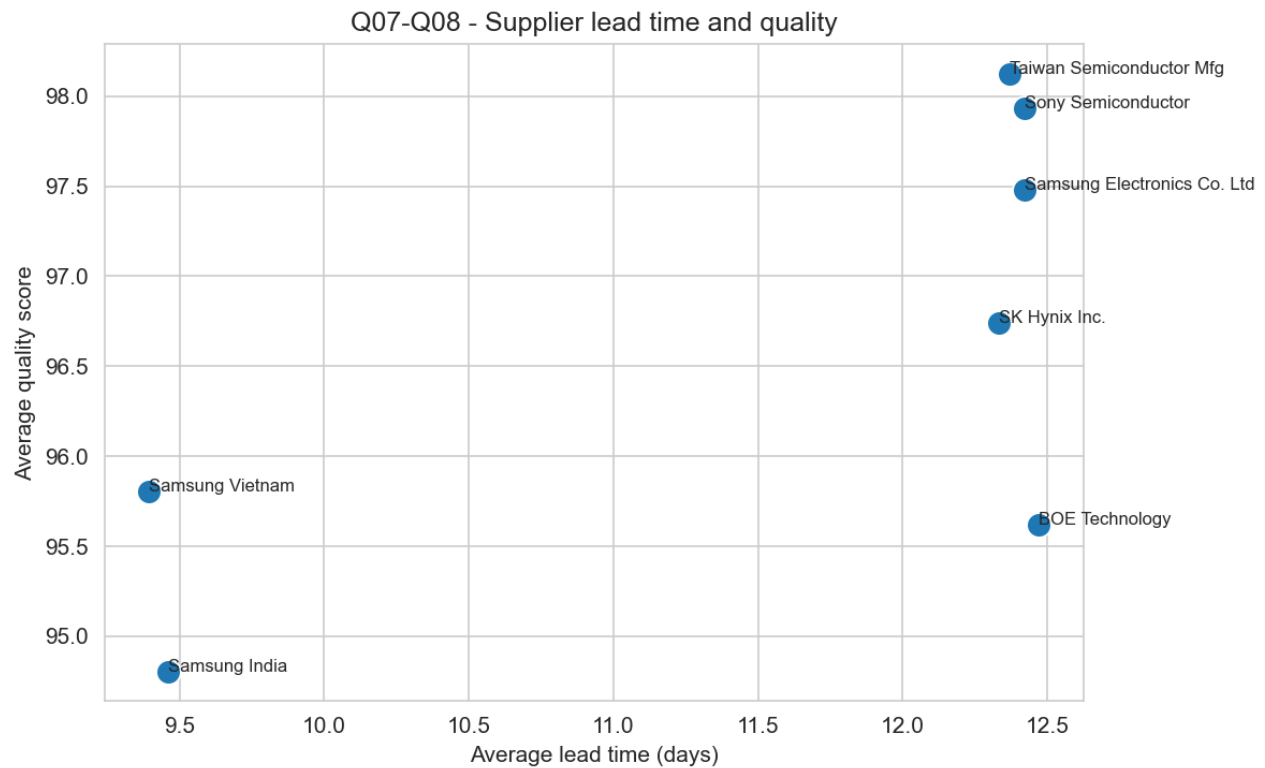
What this means: Use supplier lead time in purchase planning and review whether slower suppliers support products that are operationally important.

Supplier quality and speed

Q08 Is supplier quality related to procurement lead time?

At a glance

The correlation was **0.16**, a weak positive relationship.



What the chart shows

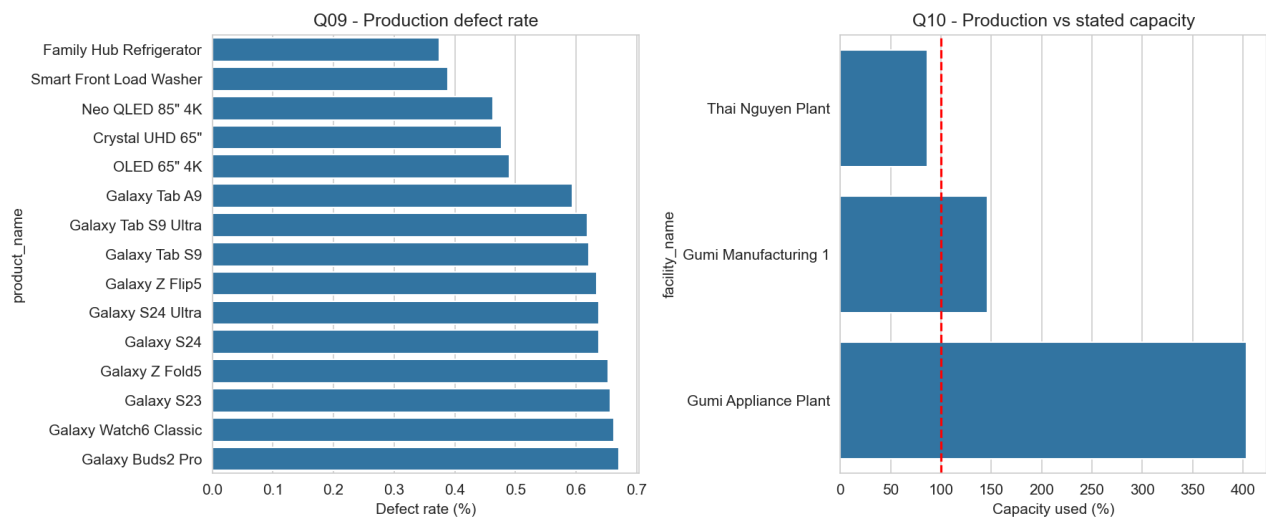
The scatter plot places each supplier by average lead time and average quality. The points do not show a strong trade-off in this dataset, so quality and speed should be reviewed separately rather than assuming one determines the other.

What this means: Build a supplier scorecard with both measures and investigate individual outliers instead of relying on one overall ranking.

Production quality

Q09 Which products have the highest production defect rates?

At a glance
Galaxy Buds2 Pro had the highest recalculated defect rate at 0.670%.



What the chart shows

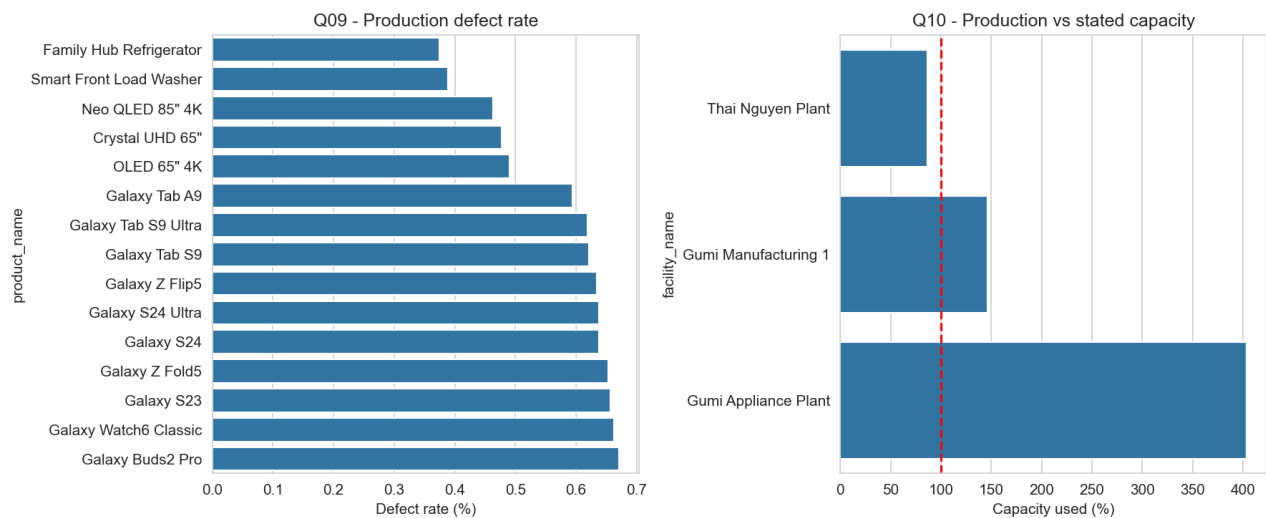
The chart compares defective units with total production through a recalculated rate. Small rate differences can still matter when production volumes are large.

What this means: Review the highest-rate products with production teams and monitor the defect rate after corrective actions.

Capacity validation

Q10 Does production output exceed stated facility capacity?

At a glance
Gumi Appliance Plant used 403.52% of stated annual capacity.



What the chart shows

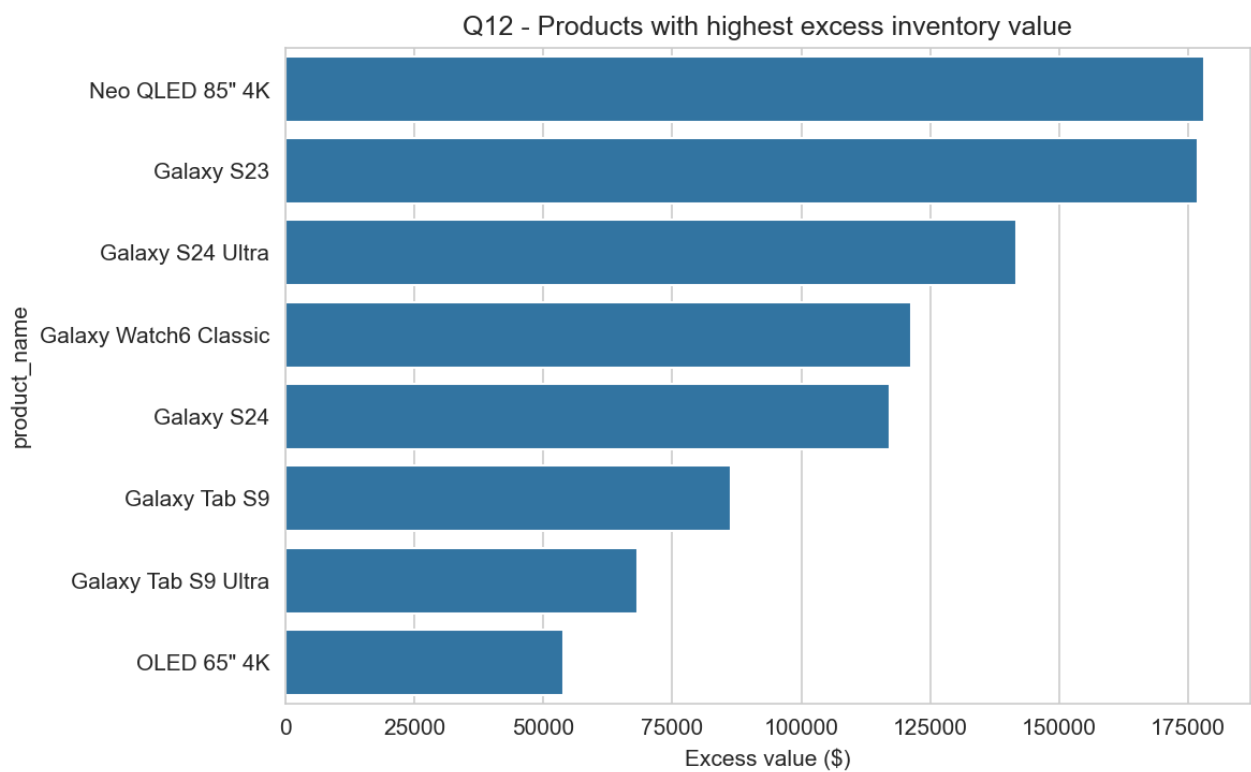
The capacity chart compares recorded production output with the annual capacity field. A result above 100% is a data-quality or definition signal, because the source may combine multiple periods or use capacity differently from the report.

What this means: Validate the capacity definition and reporting period before using this metric for investment or scheduling decisions.

Inventory replenishment risk

Q11 Which product-facility snapshots are below reorder point?

At a glance
14 of 48 latest product-facility snapshots were below reorder point.



What the chart shows

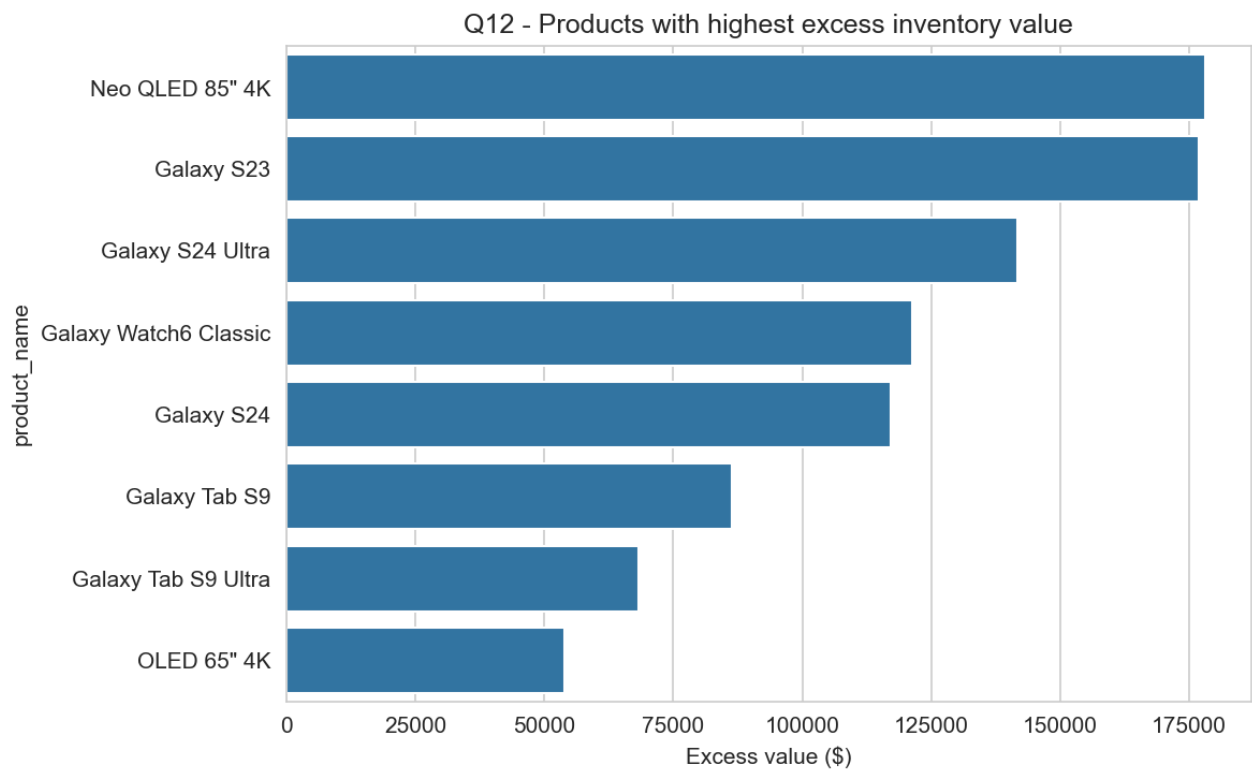
The latest inventory snapshot compares stock levels with reorder points. Below-reorder combinations may require replenishment or a transfer, while the result should not be interpreted as a full service-level forecast.

What this means: Prioritize the most severe below-reorder combinations and check whether nearby facilities have surplus stock.

Excess inventory

Q12 Which products hold the most excess inventory value?

At a glance
Neo QLED 85" 4K had \$178,034 in excess inventory value.



What the chart shows

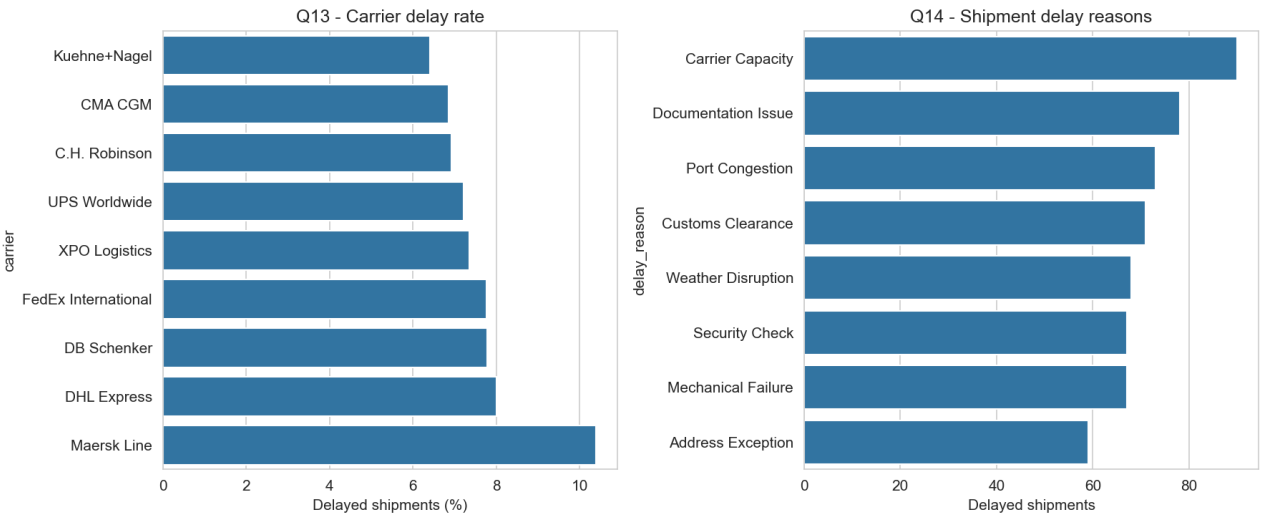
The chart estimates excess value as units above reorder point multiplied by catalog unit cost. It highlights where cash may be tied up above the current target level.

What this means: Review demand, transfers, and markdown plans for high-value excess products before placing additional purchase orders.

Carrier reliability

Q13 Which carriers have the highest recorded shipment delay rates?

At a glance
Maersk Line had the highest recorded delay rate at 10.39%.



What the chart shows

The carrier chart compares the share of shipments marked Delayed. Shipment counts should be considered with the percentage because a rate based on few shipments is less stable.

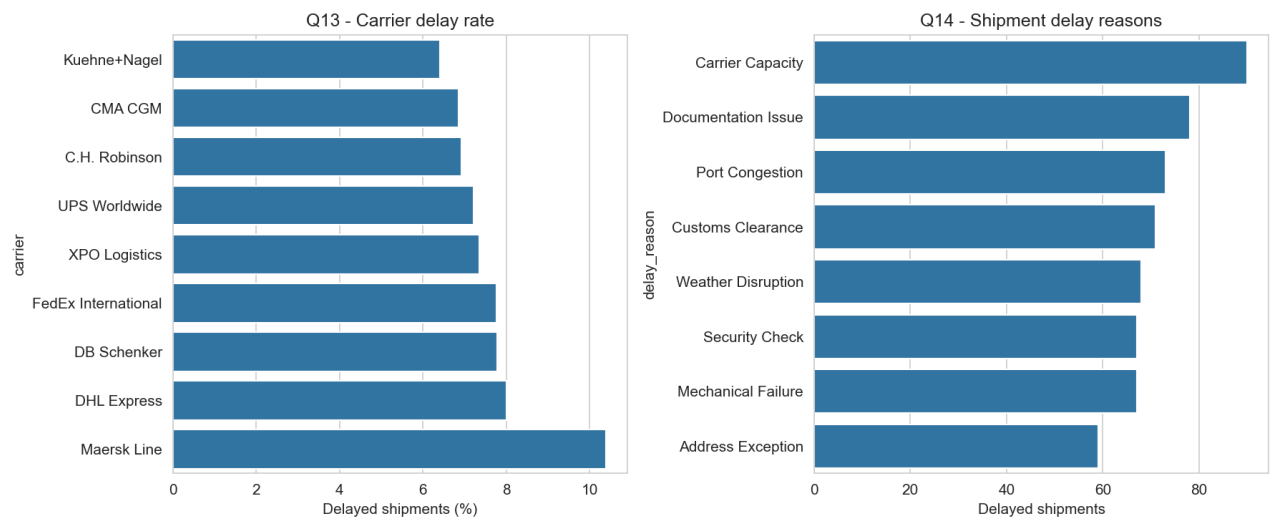
What this means: Use carrier delay rate as a review signal, then compare lanes, products, and shipment volumes before changing carrier allocations.

Delay reasons

Q14 What are the main shipment delay reasons?

At a glance

Carrier Capacity was the most common reason with 90 cases.



What the chart shows

The reason chart ranks the recorded explanations among delayed shipments. It turns a general delay KPI into possible operational workstreams such as capacity planning, documentation, or customs processes.

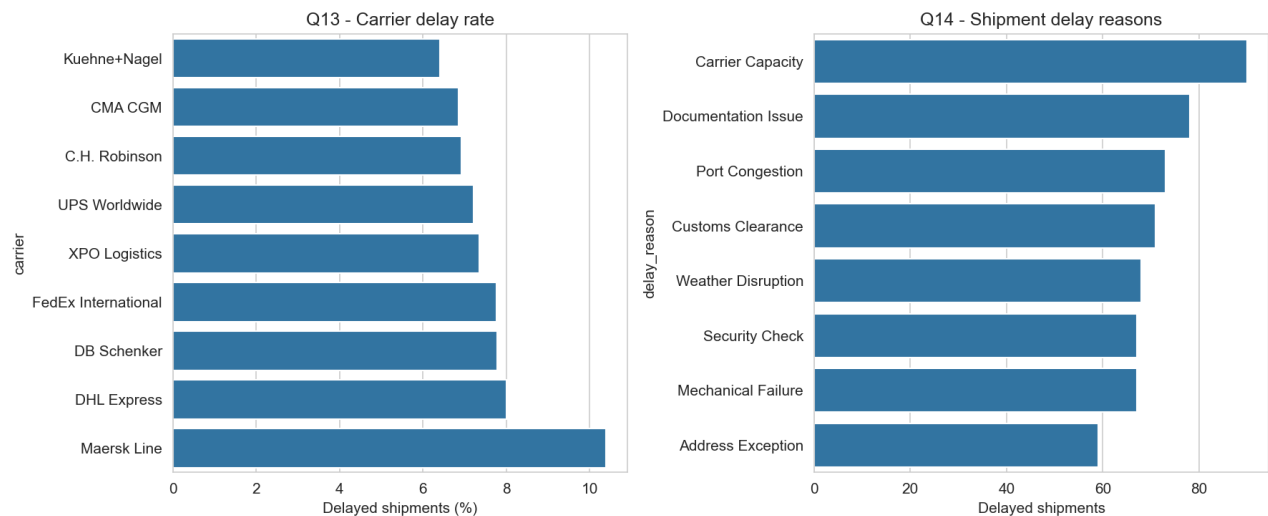
What this means: Assign an owner to the leading delay reason and track its monthly share after the process change.

Freight cost comparison

Q15 Is average freight cost different for delayed shipments?

At a glance

Average shipping cost was \$2,482 for delayed shipments and \$2,598 for non-delayed shipments.



What the chart shows

This simple comparison does not show a higher average cost for delayed shipments in the supplied data. Cost is affected by weight, route, product, and carrier mix, so it should not be used alone to judge reliability.

What this means: Create a carrier scorecard that combines cost, delay rate, shipment volume, and route mix.